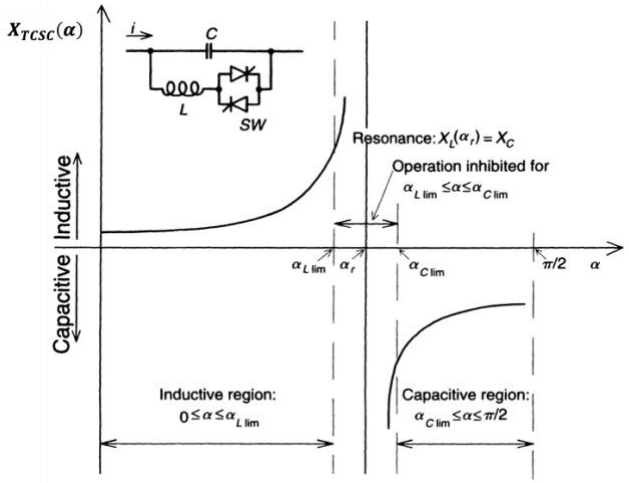
Answer ALL Questions

Question 1 on Flexible AC Transmission System (FACTS)

1. a) A thyristor-controlled series capacitor (TCSC) typically consists of a fixed capacitor in parallel with a thyristor-controlled reactor. Sketch the effective reactance offered by such a TCSC if the thyristor firing angle (α) is varied between 0° and 180°. Label the relevant points and range of interest on the plot.

[2]

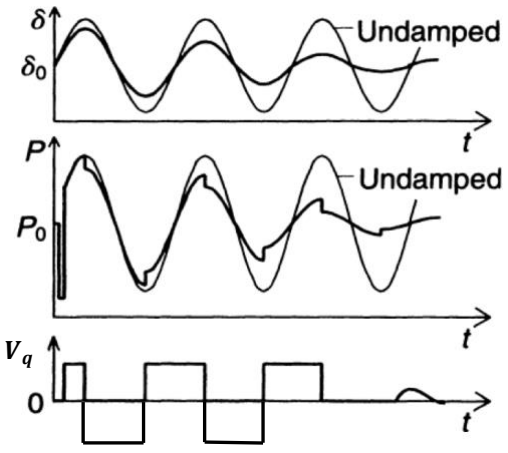


[1 mark each for the figure and correct labelling]

(bookwork – can be answered based on the material in the lecture notes)

b) Explain how a Static Synchronous Series Compensator (SSSC) installed on a transmission line can exercise power oscillation damping (POD). State the POD control law for the SSSC operating in bang-bang mode and support your explanation by sketching the typical dynamic variation of power angle (δ) across the line, the active power flow (P) through it and modulation of the control parameter of the SSSC. Include two traces in each plot showing the cases with and without power oscillation damping control through SSSC.

[3]



Model Answers

Control law

$$V_q = \begin{cases} V_{q-max} \text{ if } \dfrac{d\delta}{dt} > 0 \\ -V_{q-max} \text{ if } \dfrac{d\delta}{dt} < 0 \end{cases}$$

- With V_{q-max}, effective line impedance is minimum facilitating power transfer to reduce rate of increase of δ. When the oscillating generator (or a group of generators i.e. an area) accelerates i.e. the power angle increases, the power transmitted must increase to compensate for the excess power input to the line.

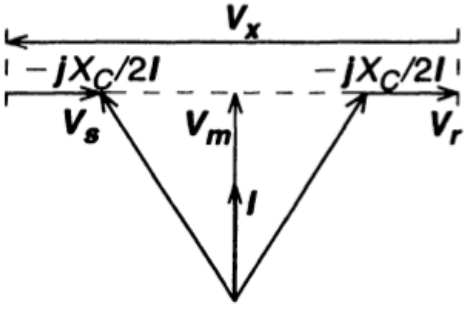
- With -V_{q-max}, effective line impedance is maximum opposing power transfer to reduce rate of decrease of δ. When the oscillating generator (or a group of generators i.e. an area) decelerates i.e. the power angle decreases, the power transmitted must decrease to balance insufficient power input to the line.

[1 mark each for the figure, control law and explanation]

(Interpretation – a similar example is provided in the lecture notes but for TCSC. Well-prepared students should be able to extend that to the SSSC case by identifying the range and influence of its control parameter on powerflow. This question is adapted to remove the factual recall elements in the view of remote 'open book' exam)

c) Draw the phasor diagram for a series compensated line with the line reactance (X) and capacitive series compensation (X_C), both split into two equal halves and assuming the voltages at the sending and receiving ends of the line to be equal in magnitude. Show the voltages at the sending end, receiving end and the midpoint of the line, voltage drops and current on the phasor diagram. Neglect line resistance and shunt charging.

[2]



(bookwork – can be answered from the material in lecture notes)

d) Using the phasor diagram in part (b), derive the expression for the reactive power (Q_C) exchanged with the series compensator in terms of the power angle (δ) across the line, line reactance (X), compensation level (k) and the voltage magnitude (V) at both ends of the line. Neglect line resistance and shunt charging.

[2]

Model Answers

$$|I| = \frac{|V_S - V_R|}{X(1-k)} = \frac{2V \sin\frac{\delta}{2}}{X(1-k)}$$

$$Q_C = |I|^2 X_C = |I|^2 k X_C = \frac{4V^2\left(\sin\frac{\delta}{2}\right)^2}{X^2(1-k)^2} kX = \frac{2V^2 k}{X(1-k)^2}(1-\cos\delta)$$

[1 mark for each step]

(deduction from the expressions provided in lecture notes)

e) A TCSC is installed on a transmission line with reactance X=0.25 p.u and negligible resistance and shunt charging. The voltages at both ends of the line are maintained at 1.0 p.u. Determine the following neglecting line resistance and shunt charging:

i) compensation level (k in %) of the TCSC required to double the maximum active power transfer limit of the line compared to that without the TCSC.

[2]

$$\frac{P'_{max}}{P_{max}} = \frac{X}{X(1-k)} = 2$$

$$k = 0.5 \text{ or } 50\%$$

ii) reactive power exchanged with the TCSC at the point of maximum possible active power transfer.

[2]

$$\delta_{max} = 90°$$

$$Q_C = \frac{2V^2 k}{X(1-k)^2}(1-\cos\delta_{max}) = 16.0 \text{ p.u.}$$

iii) critical clearing angle (in degrees) with zero compensation level ($k = 0$) of the TCSC for a power flow of 2.5 p.u. considering the same line reactance in pre- and post-fault conditions and zero power transfer during the fault.

[3]

$$P_{max0} = \frac{1}{X} = 4.0 \text{ p.u.}$$

$$P = 2.5 \text{ p.u.}$$

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max0}}\right) = 38.68°$$

Model Answers

$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 70.19°$

[1 mark for each step]

iv) critical clearing angle (in degrees) with $k = 0.25$ for a power flow of 2.5 p.u. considering the same line reactance in pre- and post-fault conditions and zero power transfer during the fault.

[2]

$k = 0.25$

$P_{max1} = \dfrac{1}{X(1-k)} = 5.33$ p.u.

$\delta_0 = \sin^{-1}\left(\dfrac{P}{P_{max1}}\right) = 27.95°$

[1 mark]

$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 82.42°$

[1 mark]

v) the allowable pre-fault power flow with $k = 0.25$ if the critical clearing angle were to be the same as in part (iii)

[2]

Under the condition specified in the problem, the critical clearing angle (δ_{cr}) depends only on the initial power angle (δ_0) which depend on $\frac{P}{P_{max}}$

In part (iii)

$\dfrac{P}{P_{max}} = \dfrac{2.5}{4} = 0.625$

[1 mark]

If P' is the power transfer possible with the same critical clearing angle (δ_{cr}) as in part (iii)

$\dfrac{P'}{P_{max1}} = 0.625 \rightarrow P' = 0.625 \times P_{max1} = 3.33$ p.u.

[1 mark]

(problem solving – use of standard formulae, deductions and interpretation)

Model Answers

Question 2 on System Operation with Low Inertia

2. a) Battery storage systems can provide frequency response much faster than thermal generators, therefore providing a service called Enhanced Frequency Response (EFR) in the UK. EFR reduces the need for the regular frequency response (R), following this rule :

$$\left(\frac{H}{f_0} - \text{EFR} \cdot \frac{\text{T}_{\text{EFR}}}{4 \cdot \Delta f_{\max}}\right) \cdot R \geq \frac{(P_{\text{L}} - \text{EFR})^2 \cdot \text{T}_{\text{d}}}{4 \cdot \Delta f_{\max}}$$

Nomenclature

H: system level of inertia (Units: MW · s).
R: amount of frequency response scheduled (Units: MW).
EFR: amount of EFR scheduled (Units: MW).
P_{L}: value of the generation outage we want to secure against (Units: MW).
f_0: nominal electric frequency of the power grid (Units: Hz).
$\Delta f_{\max}$: maximum admissible frequency deviation to avoid load shedding (Units: Hz).
T_{d}: Delivery-time of frequency response (Units: s).
T_{EFR}: Delivery-time of EFR (Units: s).

Assumptions:
1. The cost of frequency response (R): £5/MW,
2. The largest generation unit in the UK: 1.8 GW,
3. Other numerical values: f_0 = 50 Hz, Δf_{max} = 0.8 Hz, T_d = 10 s, T_{EFR} = 0.5 s.

i) If the system inertia is 150GWs, i.e. H = 150 GWs, there is a provider to offer 200MW of EFR, what is the maximum bid price (£/MW) that this provider can choose in order to be selected by the system operator. Assume that the aim of the system operator is to minimize the total cost?

[4]

1) Without EFR

R=3375MW, EFR =0,
[1 marks]

2) With EFR
R=2694MW, EFR=200MW,
[1 marks]

cost 1= 3375*5
cost 2= 2694*5 + 200 * price

In order to achieve cost1>=cost2, the maximum price is 17.025 .
[2 marks]

i) Discuss how the maximum bid price from the provider, found in i), will change if the capacity of the largest generation loss (P_L)increases?

[3]

Model Ans

The increasing capacity of the largest generation loss increases the maximum bid price for the provider.
[3 marks]

iii) Assume the system operating condition as in i) and the following:

Wind output	13,000 MW
Demand	25,000 MW
Inflexible generation	
• Must run (e.g. nuclear)	8,400 MW
• Maximum Frequency Response (R)	0 MW
CCGT units	
• Rated output	500 MW
• Minimum Stable Generation	300 MW
• Maximum Frequency Response (R)	100 MW

If the system operator purchased this 200 MW EFR, how much wind curtailment can be reduced?

[5]

1) Without EFR

R=3375MW,
Minimum Online CCGT = 34
Totol Generation = 13000 (Wind) + 8400 (Nuclear) + 10200 (CCGT) = 31600 MW
Wind Curtialment = 31600- 25000=6600 MW
[2 marks]

2) With EFR

R=2694MW,
Minimum Online CCGT = 27
Totol Generation = 13000 (Wind) + 8400 (Nuclear) + 8100 (CCGT) = 29500 MW
Wind Curtialment = 29500- 25000=4500 MW

[2 marks]

Total reduction of wind curtailment = 6600 – 4500 = 2100 MW

[1 marks]

b) Answer the following questions related to the implementation of frequency support services in renewable generation systems:

i) A wind turbine with a power rating of 6 MW is controllerd to emulate a synthetic inertia of H=4s. Determine how much the power order would change (ΔP) during a frequency excursion with a RoCoF of -1 Hz/s in a 50 Hz power system.

Model Answers

[3]

The swing equation can be used to find out how much additional power a generation unit will produce during a deceleration caused by a frequency excursion.

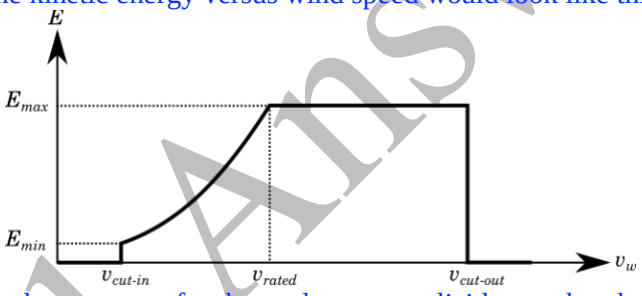
This gives:

$$\Delta P = -\frac{2H}{f_{nominal}} S_{base} \frac{df}{dt} = -\frac{2 \cdot 4\text{s}}{50\text{Hz}} \cdot 6\text{ MW} \cdot (-1\text{Hz} \cdot \text{s}^{-1}) = 0.96\text{ MW} = 0.16\text{ pu}$$

i) Draw the approximate graph of the kinetic energy stored in a wind turbine versus the wind speed. Explain the shape of the curve.

[2]

BOOKWORK. The turbine is at rest for wind speeds below cut-in or above cut-out. The turbine will spin at a speed that is linearly proportional to the wind speed under the Maximum Power Point Tracking region when the output power yield is below rated power. The turbine will operate at constant speed at any wind speed between rated speed and cut-out. The kinetic energy of the turbine is proportional to the square of the rotating speed, therefore the kinetic energy versus wind speed would look like this:



While it is not strictly necessary for the students to explicitly say that the energy will increase with the square of the wind speed in the MPPT region, they are expected to produce a graph similar to the one shown below and to justify the approximate shape of the curve considering the behavior of the turbine under the different ranges of wind speed.

i) Consider a wind turbine with an approximate moment of inertia of I_G=3.8·10$_6$ kg·m^{-2} rotating at ω_T=20 min^{-1}. Calculate how much additional power (ΔP constant) the turbine would be able to deliver to support the grid for a period of 2 seconds if the final speed of the turbine was 2% lower

[3]

The kinetic energy can be calculated as: $E = \frac{1}{2} I_G (\omega_T)^2$. The initial speed is of 20 min^{-1} which is 2.1 rad·s^{-1}, and 98% of this gives 2.06 rad·s^{-1}. Calculating the kinetic energy for these speeds one gets 8.38 MJ and 8.06 MJ respectively. This makes the reduction of 2% of the energy be equivalent to 0.32 MJ. If this energy was delivered at a constant power set point over a period of 2s, the average power would be of 160 kW.

Model Answers

3. **Question 3 on Power System State Estimation**

3. a)

i) What is the purpose of static state estimation in power systems?

[4]

State estimation is required as the operator needs to know network operational status at any time. The operational status means – how much is the flow in any branch, node voltage magnitudes, how many circuits are under operation and what is the percentage of their loading, whether network voltages and flows are within the limit set by grid operation code etc. The answer to these questions cannot be obtained with measurement only as not every node and branch are measured, neither all the quantities can be easily measured. There are errors associated with measurements also. It is required to obtain an accurate value of the quantities to synthesise other quantities those are not measured yet important for the operator to know. All these circumstances and necessity lead to the requirement of state estimation.

ii) What quantities are estimated in static state estimation?

[3]

- RMS node voltage magnitude
- node angle
- voltage control setting such as tap in transformer, switch reactor/capacitor

iii) List the major steps in static state estimation.

[3]

- topology processing
- observability analysis
- state estimation solution and
- bad data processing

b)

i) List three important challenges in distribution system state estimation.

[3]

- inadequate measurement
- poor quality measurement
- unbalance operation of the network

ii) List three main statistical criteria for assessing the effectiveness of an algorithm in distribution system state estimation.

Model Answers

[3]

- quality
- consistency
- bias

iii) Explain one of the statistical criteria for assessing the effectiveness of a state estimation algorithm in power distribution context.

[4]

One of the following three is expected

Bias: For the estimator to be unbiased, the mean error in the estimates should be zero. This means take a large number of scenarios around a base case – the voltage and or angle estimated error should be unbiased around the average error. Ideally, the average error is zero.

Or

Consistency: For consistency, the normalized mean square estimation error should be within confidence bound determined by chi-square test (under Gaussian assumption of error). This when connected with respect to the cardinality of the problem, it should be close to the number of states being estimated

Or

Quality: Expressed as the inverse of the trace of the error covariance matrix. I am expecting the student to write the state error covariance matrix and why the trace is better than determinant from the solvability point of view.

Model Answers

Question 4 on Integrated Energy Systems

4. **Part I**

a) List and briefly describe the key challenges associated with integration of wind generation in the future UK system, considering both short term operation and long term planning time horizons.

[4]

Two key points should be made.

(a1) Wind generation will displace energy produces by conventional (gas and coal) generation but the ability to displace capacity is limited – capacity credit of wind is limited to about 10%.

[2]

(a2) In the very short time scales (seconds-minutes) the dominant driver for balancing is the need to manage potential loss of conventional plant, which is increasing challenging due to reduced inertia driven by level of wind penetration (response requirement). Furthermore, reserve requirements will be driven by with the wind forecast uncertainty over time scales of minutes to several hours, which will require scheduling of reserve, generally by part loading conventional generation and applying flexibility technologies, such demand side response and storage.

[2]

b) What are the advantages and disadvantages of thermal energy storage compared to Electricity Storage?

Cost of TES is much lower than ES.
Heat loss, the longer heat is stored, the more heat is lost.
TES can only provide flexibility when there is link between electricity and heat system.

[3]

Part II

Considering the electrification of heat sector, heat systems could provide flexibility to the electricity system and support more cost-effective integration of variable wind generation, through thermal energy storage (TES). Consider a system with capacity of wind generation of 35GW, nuclear generation 12.4GW, with the rest of the load being supplied by a number of CCGT plantsm, which are all the same type with same characteristics. Electricity demand includes non-heat demand and electrified heat demand. Operating reserve is required to deal with uncertainties in wind generation production, which is managed by part-loading CCGT.

Key characteristics of the CCGT plant are:

Capacity: 550 MW
Minimum Stable Generation (MSG): 300 MW
Start-up time: 1 hour (it takes 1 hour for a CCGT plant to start-up)

System operational conditions and reserve requirements are given in the table below, considering 2 consecutive hours of system operation.

Model Answers

	1st hour	2nd hour
Wind available (MW)	28,800	14,400
Nuclear (MW)	12,400	12,400
Electricity Demand (MW) - Non-heat demand - Electrified heat demand	51,000 36,000 15,000	51,000 36,000 15,000
Reserve requirement (MW)	14,400	9,600

a) Assuming that TES is not available, answer the following questions:

i) How many CCGT plants must be online in the 2nd hour?

62 CCGTs
In the 2nd hour,
Power from CCGT = 51000MW - 14400MW - 12400MW= 24200MW
Reserve from CCGT = 9600MW
Number of CCGT = (24200 + 9600)/550 = 61.5
Therefore, 62 CCGTs are need in the 2nd hour.

[3]

ii) How much wind would be curtailed in the 1st hour (consider CCGT start-up time)?

Considering the start-up time of CCGT, at least 62 CCGTs must be online in the 1st hour.
Minimum power from CCGT = 62*300MW = 18600MW
Reserve can be provided = 62*(550MW – 300MW) = 15500MW > 14400MW
Wind power = 51000MW – 18600MW – 12400MW = 20000MW
Wind curtailment = 28800MW - 20000MW = 8800MW

[3]

b) Consider that TES is available. The efficiency of TES is 95%, i.e., 5% of stored thermal energy is lost per hour.

i) Calculate the minimum (power equivalent) capacity of TES that would be needed to ensure no wind is curtailed in the 1st hour.

[3]

Number of CCGT to provide reserve = 14400MW / (550MW – 300MW) = 58
Minimum Generation from CCGT = 58*300MW = 17400MW
Surplus power = 17400MW + 12400MW +28800MW – 51000MW = 7600MW
Therefore, 7600MW equivalent TES is needed.

ii) How many CCGTs have to be online at the 2nd hour?

[2]

Power from CCGT = 51000MW – 7600MW*95% - 14400MW – 12400MW = 16980MW

Model Answers

Reserve from CCGT = 9600MW
Number of CCGT = (16980MW +9600MW) / 550MW = 49

iii) If CO_2 emission associated with the operation of CCGT is 0.4ton/MWh, how much CO_2 is saved by using TES?

[2]

CO2 savings = 7600MW*1h*95%*0.4ton/MWh = 2888ton

Model Answers